

Agricultural Index Insurance: An Optimization Approach

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August 20, 2025

The Problem of Agricultural Risk

- Farmers face a lot of risk, and the lack of risk management tools forces them to use coping strategies that hurt their long term welfare.
- Traditional insurance is prohibitively costly in most developing countries due to lack of data and high verification costs.
- Moral hazard, adverse selection, and the presence of large covariate shocks make the problem of agricultural insurance especially hard.

A Proposed Solution: Index Insurance

- In index insurance, an index (or statistic) is created using easily observable quantities (e.g. rainfall), and it is used to determine whether the insured party suffered an adverse event.
- If the index falls below a pre-determined threshold, the insurer automatically issues out payments to the insured.
- This allows the insurer to circumvent the issue of verification and moral hazard, since the actions of individual farmers cannot affect the index.

Index Insurance in Practice

- Since it was first proposed, index insurance programs have been implemented in many countries including India, Mexico, Tanzania, Malawi, Kenya, and many others (Jensen and Barrett (2017)).
- Today, tens of millions of farmers worldwide are covered by index insurance programs (Greatrex et al. (2015)).
- However, in most of these cases, the insurance has to be heavily subsidized by governments due to high cost and low demand (Greatrex et al. (2015)).

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Index Insurance: General Context

- Index insurance is a named-peril insurance.
- It is often tied to access to credit.
- Offered at different levels (micro level, meso level, and macro level)
- Often based on public-private partnerships. Research institutions usually design the product, and insurance companies (or governments) sell it.

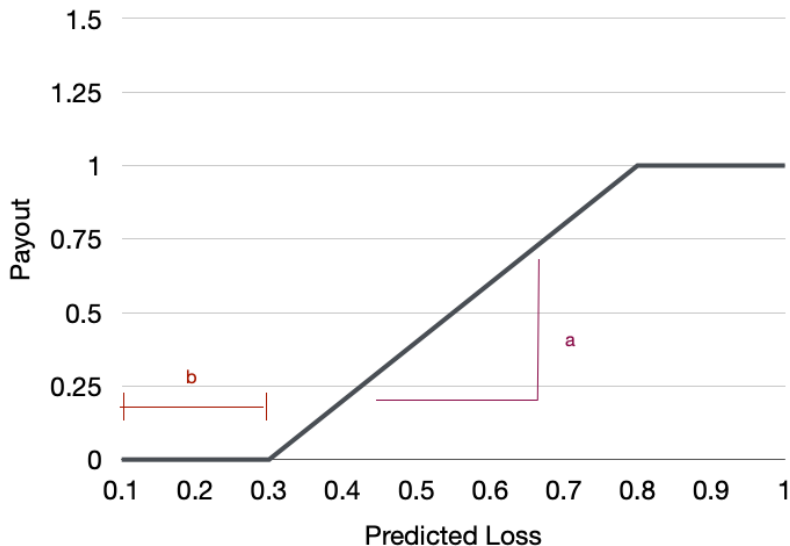
Index Insurance: Definition and Parameters

- Index insurance uses a signal, θ , to predict agricultural loss, $\hat{\ell}(\theta)$
- Contract form: $I(\hat{\ell}(\theta)) = \min \left\{ \max \left\{ a\hat{\ell}(\theta) - b, 0 \right\}, 1 \right\}$, a, b are the contract parameters.

Reasoning for piecewise linear contracts

- Research has consistently found that contract comprehension is key for take up (Cole et al. (2013); Ahmed, McIntosh, and Sarris (2020))
- Deductible insurance is theoretically optimal (Arrow (1974))
- Widely used in practice (Gamage (2021); Bank (2011))

Example of Index Insurance Contract



Motivation for multi-zone optimization

Insurer's expected costs for insuring its entire portfolio are:

$$C = \mathbb{E} \left[\sum_z s_z I_z(\hat{\ell}_z) \right] + c_\kappa R \left(\sum_z s_z I_z(\hat{\ell}_z) \right)$$

Charges per-unit premium of:

$$\pi_z = \mathbb{E} \left[I_z(\hat{\ell}_z) \right] + c_\kappa \frac{\beta_z}{s_z} R \left(\sum_z s_z I_z(\hat{\ell}_z) \right)$$

Where $R(\cdot)$ is a risk measure capturing capital requirements, s_z is the number of insured units in zone z , and β_z is the share of the capital costs paid by zone z .

Insurer's Optimization Problem

Insurer wants to find $\mathbf{a} = (a_1, \dots, a_Z)$, $\mathbf{b} = (b_1, \dots, b_Z)$ that will maximize overall utility.

$$\begin{aligned} \max_{\mathbf{a}, \mathbf{b}} \quad & \mathbb{E} \left[\sum_z s_z u \left(w_0 + 1 - \ell_z - \pi_z + I_z(\hat{\ell}_z) \right) \right] \\ \text{s.t.} \quad & \pi_z = \mathbb{E} \left[I_z(\hat{\ell}_z) \right] + c_\kappa \frac{\beta_z}{s_z} R \left(\sum_z s_z I_z(\hat{\ell}_z) \right) \end{aligned}$$

FOC's for a_z :

$$\frac{\partial I_z}{\partial a_z} - \mathbb{E} \left[\frac{\partial I_z}{\partial a_z} \right] = c_\kappa \beta_z R' \left(\sum_z s_z I_z(\hat{\ell}_z) \right) \frac{\partial I_z}{\partial a_z}$$

Insurer's Optimization in Single Zone Case

If designing the contracts independently, the insurer's optimization problem becomes:

$$\begin{aligned} \max_{a,b} \quad & \mathbb{E} \left[u \left(w_0 + 1 - \ell_z - \pi_z + I_z(\hat{\ell}_z) \right) \right] \\ \text{s.t.} \quad & \pi_z = \mathbb{E} \left[I_z(\hat{\ell}_z) \right] + c_\kappa \frac{\beta_z}{s_z} R \left(s_z I_z(\hat{\ell}_z) \right) \end{aligned}$$

FOC's for a_z :

$$\frac{\partial I_z}{\partial a_z} - \mathbb{E} \left[\frac{\partial I_z}{\partial a_z} \right] = c_\kappa R' \left(s_z I_z(\hat{\ell}_z) \right) \frac{\partial I_z}{\partial a_z}$$

Comparing FOC's

Multi-zone Optimization FOC's:

$$\frac{\partial l_z}{\partial a_z} - \mathbb{E} \left[\frac{\partial l_z}{\partial a_z} \right] = c_\kappa \beta_z R' \left(\sum_z s_z l_z(\hat{\ell}_z) \right) \frac{\partial l_z}{\partial a_z}$$

Single-zone Optimization FOC's:

$$\frac{\partial l_z}{\partial a_z} - \mathbb{E} \left[\frac{\partial l_z}{\partial a_z} \right] = c_\kappa R' \left(s_z l_z(\hat{\ell}_z) \right) \frac{\partial l_z}{\partial a_z}$$

FOC's when $R(\cdot) = sd(\cdot)$ and $|Z| = 2$

Multi-zone Optimization FOC's:

$$\hat{\ell}_1 - \mathbb{E} [\hat{\ell}_1] = c_\kappa \beta_1 \hat{\ell}_1 \frac{\left(a_1 \text{Var}(\hat{\ell}_1) + a_2 \text{Cov}(\hat{\ell}_1, \hat{\ell}_2) \right)}{\sqrt{a_1^2 \text{Var}(\hat{\ell}_1) + a_2^2 \text{Var}(\hat{\ell}_2) + 2a_1 a_2 \text{Cov}(\hat{\ell}_1, \hat{\ell}_2)}}$$

Single-zone Optimization FOC's:

$$\hat{\ell}_1 - \mathbb{E} [\hat{\ell}_1] = c_\kappa \hat{\ell}_1 sd(\hat{\ell}_1)$$

Index Insurance Literature

- **Impact of Index Insurance:** Overall, there is evidence that index insurance reduces reliance on detrimental risk-coping strategies, increases investment, and leads to riskier, but more profitable production decisions (Jensen and Barrett (2017); Cole et al. (2013); Mobarak and Rosenzweig (2013); Karlan et al. (2014)).
- **Demand for Index Insurance:** Demand for index insurance tends to be low and highly price sensitive (Jensen and Barrett (2017); Cole et al. (2013); Cai, De Janvry, and Sadoulet (2020), Casaburi and Willis (2018)).
- **Design of Index Insurance:** There has been relatively little research studying the design of index insurance. The method developed by Chantarat et al. (2013) is the most commonly used in academic publications (Jensen, Stoeffler, et al. (2019); Flatnes, Carter, and Mercovich (2018)). More recently, Chen et al. (2023) used a NN approach for contract design.

Baseline Method: Chantarat et al. (2013)

Developed by Chantarat et al. (2013), used in Kenya Index Based Livestock Insurance (IBLI) program. Consist of two steps:

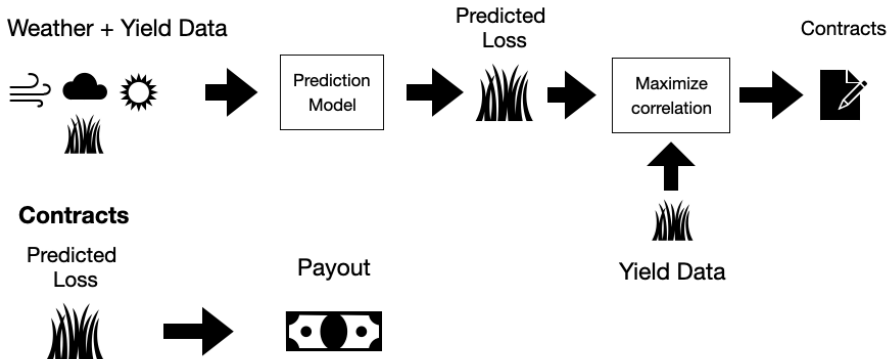
- 1 Train model to predict loss
- 2 Choose strike value, b , that maximizes correlation between payouts and losses

Pros: simple, interpretable

Cons: can't incorporate constraints, doesn't take into account premium

Baseline Method Flowchart

Flowchart of baseline method



NN Approach: Chen et al. (2023)

Train neural network to solve the following optimization problem:

$$\begin{aligned} \max \quad & \mathbb{E}[u(w_0 + 1 - \ell + I - \pi)] \\ \text{s.t.} \quad & \pi = \lambda \mathbb{E}[I] \\ & \pi \leq \bar{\pi} \end{aligned}$$

Use CARA utility function:

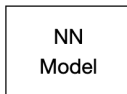
$$\begin{aligned} u(w) &= \frac{-1}{\alpha} e^{-\alpha w} \\ \alpha &> 0 \end{aligned}$$

NN Approach: Cont'd

- Use a feed forward NN with 3 hidden layers of size 64-64-16.
- Evaluate their method using corn yield data from the US Midwest.
- **Pros:** Directly maximizes utility, admits price constraints.
- **Cons:** Not interpretable. hard to adjust, large data requirements

Flowchart of Chen et al 2023 Method

Weather + Yield Data



Contracts



Contracts

Weather



Payout



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Our Approach: Overview

Our approach is most similar to the baseline method. We predict-then-design. The steps are:

- 1 Train a traditional ML model to predict loss
- 2 Use model predictions and ground truth to design contracts

This yields simple and interpretable contracts.

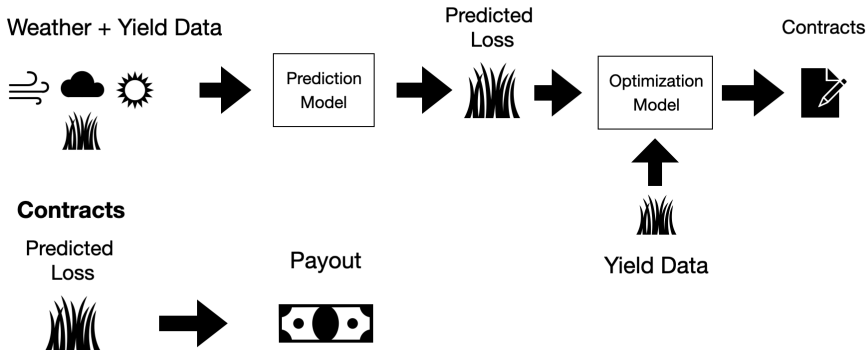
- We opt for a "predict-then-optimize" approach.
- We use specialized time-series feature extraction algorithms and traditional ML algorithms (e.g. Random Forest, Gradient Boosting, Support Vector Machines) to build a loss prediction model.
- We then use model predictions and actual losses to design contracts to maximize utility.

Method Comparison

	Baseline	Chen	VMX
Simple contracts	✓	✗	✓
Adjustable contracts	✓	✗	✓
Price constraints	✗	✓	✓
Payout frequency constraints	✗	✗	✓
Different premiums defn's	✗	✓	✓
Multi-zone contract design	✗	✗	✓

Our Method Flowchart

Flowchart of our Method



Prediction: Problem Overview

- For each plot i , we have several time-series of different weather variables, sampled at potentially different frequencies:

$$\begin{bmatrix} x_{i,1} & \dots & x_{i,16} & \dots & x_{i,176} \\ y_{i,1} & y_{i,2} & \dots & y_{i,16} & \dots & y_{i,176} & \dots & y_{i,180} \\ z_{i,1} & z_{i,2} & \dots & z_{i,16} & \dots & z_{i,176} & \dots & z_{i,180} \end{bmatrix}$$

- These weather variables are rainfall, temperature, and evapotranspiration.

Feature Extraction

- Time-series classification (TSC) is different from standard problems because the data points are ordered, and the order matters. The space of potential features is also large, e.g. we can take 3-day dry spells, 5-day dry spells, number of high temperature days, etc.
- Ruiz et al. (2021) compared many different specialized TSC algorithms and they found that the RoCKET algorithm (Dempster, Petitjean, and Webb (2020)) generally performed best.
- We use RoCKET, Catch22 (Lubba et al. (2019)), and raw time-series for feature extraction.

Prediction: Full Pipeline

- **Preprocessing:** uniform scaling, quantile transformer
- **Feature extraction:** Catch22, RoCKET, raw time-series
- **Prediction algorithm:** Random Forests (RF), Gradient Boosting Regressor (GBR), Elastic Net, LASSO

Framework

Farmer wealth:

$$w = w_0 + 1 - \ell + I(\hat{\ell}) - \pi$$

Premium ¹farmer pays:

$$\pi = \mathbb{E}[I(\hat{\ell})] + c_{\kappa} K$$

$$K = \text{CVaR}_{1-\epsilon} \left(I(\hat{\ell}) \right) - \mathbb{E}[I(\hat{\ell})]$$

CRRA Utility function:

$$u(w) = \frac{1}{1-\alpha} w^{1-\alpha}$$

¹Premium definition comes from *Risk Modeling for Appraising Named Peril Index Insurance Products: A Guide for Practitioners* (Mapfumo, Groenendaal, and Dugger (2017))

Premium Principles

A premium principle is a functional that maps an insurance contract to a price (premium). There are several premium principles that are common in the actuarial literature, and our method is compatible with most of them:

Description	Definition
Expected value	$(1 + \alpha)\mathbb{E}[I]$
Standard deviation	$\mathbb{E}[I] + \alpha\sigma(I)$
Variance	$\mathbb{E}[I] + \alpha\sigma^2(I)$
Exponential	$\frac{1}{\alpha} \log \mathbb{E}[e^{\alpha I}]$
Dutch	$\mathbb{E}[I] + \beta\mathbb{E}[(I - \alpha\mathbb{E}[I])_+]$

Idealized Model

$$\max_{a,b,\pi,K} \mathbb{E} [U(w_0 + 1 - \ell - \pi + I(\theta))] \quad (1)$$

$$\text{s.t.} \quad I(\theta) = \min \left\{ \max \left\{ 0, a\hat{\ell}(\theta) + b \right\}, 1 \right\} \quad (2)$$

$$\pi = \mathbb{E} [I(\theta)] + c_{\kappa} [\text{CVaR}_{1-\epsilon_K} (I(\theta)) - \mathbb{E}[I(\theta)]] \quad (3)$$

$$\pi \leq \bar{\pi}. \quad (4)$$

Convex Relaxation

$$\max_{a,b,K,\pi} \mathbb{E} \left[U \left(w_0 + 1 - \ell - \pi + \underline{I(\theta)} \right) \right] \quad (5)$$

$$\text{s.t.} \quad \overline{I(\theta)} = \max \left\{ 0, a\hat{\ell}(\theta) + b \right\}$$

$$\underline{I(\theta)} = \min \left\{ a\hat{\ell}(\theta) + b, 1 \right\}$$

$$\pi = (1 - c_\kappa) \mathbb{E} \left[\overline{I(\theta)} \right] + c_\kappa \text{CVaR}_{1-\epsilon_K} \left(\overline{I(\theta)} \right) \quad (6)$$

$$\pi \leq \overline{\pi}.$$

Multiple Zone Case

In the multi-zone case, the problems are coupled through the required capital

$$K = \text{CVaR}_{1-\epsilon} \left(\sum_z s_z I_z(\hat{\ell}_z) \right) - \mathbb{E} \left[\sum_z s_z I_z(\hat{\ell}_z) \right]$$

The first term will be affected by the correlation of losses across zones, so the optimal contracts will vary depending on this correlation. There are also additional decision variables k_z .

Multiple Zone Model

$$\begin{aligned} \max_{a,b,K,\pi} \quad & \mathbb{E} \left[\sum_z s_z U(w_{0,z} + 1 - \ell_z - \pi_z + \underline{l}_z(\theta_z)) \right] \\ \text{s.t.} \quad & \pi_z = \mathbb{E} \left[\overline{l}_z(\theta_z) \right] + \frac{c_K}{s_z} k_z \end{aligned} \quad (7)$$

$$K = \text{CVaR}_{1-\epsilon_K} \left(\sum_z s_z \overline{l}_z(\theta_z) \right) - \mathbb{E} \left[\sum_z s_z \underline{l}_z(\theta_{z'}) \right] \quad (8)$$

$$K = \sum_z k_z \quad (9)$$

$$\pi_z \leq \overline{\pi}_z \quad (10)$$

$$\overline{l}_z(\theta_z) = \max \left\{ 0, a_z \hat{\ell}_z(\theta_z) + b_z \right\}$$

$$\underline{l}_z(\theta_z) = \min \left\{ a_z \hat{\ell}_z(\theta_z) + b_z, 1 \right\}$$

Single-zone vs Multi-zone

- $P1(z)$ - optimization program that designs contract for zone z independently of other zones.
- $P2$ - multi-zone optimization program.
- Let $P1(z)^*$ be the optimal value of $P1$ on zone z and let $P2^*$ be the optimal value of $P2$. We want to compare $\sum_z P1(z)^*$ and $P2$.
- We will show that $P2$ is a relaxation of running $P1$ for all zones and taking the sum. That is $P2^* \geq \sum_z P1(z)^*$.

SZ vs MZ: Constructing a solution

- Let $\mathbf{a}^{SZ} = (a_1^{SZ}, \dots, a_Z^{SZ})$ and $\mathbf{b}^{SZ} = (b_1^{SZ}, \dots, b_Z^{SZ})$ be the optimal values of a, b obtained from running $P1(z)$ for every zone, and let $\boldsymbol{\pi}^{SZ} = (\pi_1^{SZ}, \dots, \pi_Z^{SZ})$ and $\mathbf{K}^{SZ} = (K_1^{SZ}, \dots, K_Z^{SZ})$ be the corresponding values of π and K implied by a and b in each case.
- We denote our proposed solution for $P2$ with the superscript MZ .
- We set $\mathbf{a}^{MZ} = \mathbf{a}^{SZ}$ and $\mathbf{b}^{MZ} = \mathbf{b}^{SZ}$ and $\overline{K^{SZ}} = \sum_z s_z K_z^{SZ}$.
- Let K^{MZ} be the value of K in $P2$ implied by our proposed $\mathbf{a}^{MZ}$ and $\mathbf{b}^{MZ}$. Define $\Delta K = \overline{K^{SZ}} - K^{MZ}$. Set $k_z^{MZ} = s_z K_z^{SZ} - \frac{\Delta K}{|Z|}$.

SZ vs MZ: Feasibility

Constraints (7) and (8) hold automatically. Want to check $\sum_z k_z = K^{MZ}$.

$$\begin{aligned}\sum_z k_z^{MZ} &= \sum_z \left(s_z K_z^{SZ} - \frac{\Delta K}{|Z|} \right) \\ &= \sum_z s_z K_z^{SZ} - (\overline{K^{SZ}} - K^{MZ}) \\ &= K^{MZ}\end{aligned}$$

SZ vs MZ: Feasibility

Need to verify $\pi_z \leq \overline{\pi_z}$ for all z .

$$\begin{aligned}\pi_z^{SZ} &= \mathbb{E} \left[I_z(\hat{\ell}_z) \right] + c_\kappa K_z^{SZ} \\ &= \mathbb{E} \left[I_z(\hat{\ell}_z) \right] + \frac{c_k}{s_z} s_z K_z^{SZ} \\ \pi_z^{MZ} &= \mathbb{E} \left[I_z(\hat{\ell}_z) \right] + \frac{c_\kappa}{s_z} k_z^{MZ} \\ &= \mathbb{E} \left[I_z(\hat{\ell}_z) \right] + \frac{c_\kappa}{s_z} (s_z K_z^{SZ} - \frac{\Delta K}{|Z|})\end{aligned}$$

Constraint will be satisfied if $\Delta K \geq 0$

$$\Delta K \geq 0$$

$$\begin{aligned}\Delta K &= \overline{K^{SZ}} - K^{MZ} \\ \overline{K^{SZ}} &= \sum_z s_z K_z^{SZ} \\ &= \sum_z s_z \left[\text{CVaR}(I_z(\hat{\ell}_z)) - \mathbb{E} \left[I_z(\hat{\ell}_z) \right] \right] \\ &= \sum_z \text{CVaR}(s_z I_z(\hat{\ell}_z)) - \mathbb{E} \left[\sum_z s_z I_z(\hat{\ell}_z) \right] \\ &\geq \text{CVaR} \left(\sum_z s_z I_z(\hat{\ell}_z) \right) - \mathbb{E} \left[\sum_z s_z I_z(\hat{\ell}_z) \right] \\ &= K^{MZ}\end{aligned}$$

SZ vs MZ: Comparing Objectives

SZ Objective:

$$\sum_z P1(z) = \sum_z s_z u(w_0 + 1 - \ell_z + I_z - \pi^{SZ})$$

MZ Objective:

$$P2 = \sum_z s_z u(w_0 + 1 - \ell_z + I_z - \pi^{MZ})$$

The only difference between the objectives is π^{MZ} and π^{SZ} . Since $u()$ is monotonically increasing, and $\pi^{MZ} \leq \pi^{SZ}$ we have that $P2 \geq \sum_z P1(z)$

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Data in Index Insurance Research

- One of the biggest barriers in AI research is the lack of granular yield data for LMIC's.
- Most previous studies either use a small dataset from a pilot index insurance program in Kenya (Chantararat et al. (2013); Flatnes, Carter, and Mercovich (2018); Estefania-Salazar et al. (2024)) or use data from a rich country (Chen et al. (2023)).
- I collaborated with the Bank of Thailand and obtained access to a novel dataset of losses from natural disasters in Thailand.

Overview of Agriculture in Thailand

- Characterized by rural, smallholder farmers, with an average farm size of 5.7 acres. Average in sub-Saharan Africa, South Asia, and East Asia is ~ 4 acres.
- 76% of agriculture in Thailand is rain-fed. Globally, this ranges from 60% in South Asia to over 90% in sub-Saharan Africa.
- Consistent exposure to floods and droughts, especially in northeast region. LMIC's generally face more flood and drought risk than richer countries (Winsemius et al. (2018)).
- $\sim 30\%$ of Thai labor force is employed in agriculture.

Thai Data: Administrative Data Sources

- Thailand Department of Agricultural Extension (DOAE): This dataset contains detailed information on the planting activities of over 3.1 million Thai farmers. Information includes plant date, plot size, rice variety, etc
- Thai Government Disaster Relief Data: This data is from the current national disaster relief program. It contains information about the disaster type, losses, and disaster date.

Remote Sensing Data Sources

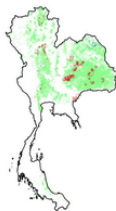
- Climate Hazards Center InfraRed Precipitation with Station data (CHIRPS): daily rainfall data 1981-present
- MOD21A1 Land Surface Temperature: daily temperature data, available 2000-present
- FLDAS Famine Early Warning Systems Network: monthly evapotranspiration data, available 1992-present

Data Summary Statistics

Characteristic	Overall N = 49,052 [†]	2015 N = 6,037 [†]	2016 N = 6,358 [†]	2017 N = 6,285 [†]	2018 N = 6,288 [†]	2019 N = 6,268 [†]	2020 N = 6,079 [†]	2021 N = 5,510 [†]	2022 N = 6,227 [†]
Loss	0.02 (0.08)	0.00 (0.03)	0.02 (0.08)	0.04 (0.11)	0.01 (0.05)	0.01 (0.07)	0.00 (0.03)	0.03 (0.10)	0.03 (0.11)
Zone									
C1	6,052 (12%)	743 (12%)	763 (12%)	757 (12%)	761 (12%)	762 (12%)	766 (13%)	737 (13%)	763 (12%)
C2	3,064 (6.2%)	366 (6.1%)	404 (6.4%)	394 (6.3%)	393 (6.3%)	396 (6.3%)	391 (6.4%)	329 (6.0%)	391 (6.3%)
C3	3,139 (6.4%)	392 (6.5%)	426 (6.7%)	405 (6.4%)	410 (6.5%)	410 (6.5%)	398 (6.5%)	294 (5.3%)	404 (6.5%)
N1	6,312 (13%)	784 (13%)	808 (13%)	805 (13%)	803 (13%)	798 (13%)	798 (13%)	718 (13%)	798 (13%)
N2	3,593 (7.3%)	451 (7.5%)	451 (7.1%)	454 (7.2%)	449 (7.1%)	451 (7.2%)	450 (7.4%)	437 (7.9%)	450 (7.2%)
N3	2,105 (4.3%)	269 (4.5%)	268 (4.2%)	268 (4.3%)	265 (4.2%)	264 (4.2%)	265 (4.4%)	241 (4.4%)	265 (4.3%)
NE1	1,294 (2.6%)	162 (2.7%)	162 (2.5%)	162 (2.6%)	162 (2.6%)	162 (2.6%)	162 (2.7%)	160 (2.9%)	162 (2.6%)
NE2	9,400 (19%)	1,178 (20%)	1,181 (19%)	1,183 (19%)	1,183 (19%)	1,179 (19%)	1,178 (19%)	1,138 (21%)	1,180 (19%)
NE3	10,546 (21%)	1,317 (22%)	1,320 (21%)	1,319 (21%)	1,319 (21%)	1,320 (21%)	1,319 (22%)	1,315 (24%)	1,317 (21%)
S1	1,025 (2.1%)	111 (1.8%)	145 (2.3%)	141 (2.2%)	140 (2.2%)	148 (2.4%)	127 (2.1%)	76 (1.4%)	137 (2.2%)
S2	2,522 (5.1%)	264 (4.4%)	430 (6.8%)	397 (6.3%)	403 (6.4%)	378 (6.0%)	225 (3.7%)	65 (1.2%)	360 (5.8%)

[†] Mean (SD); n (%)

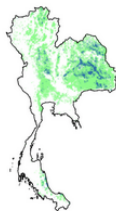
Disaster Activity Map



2015



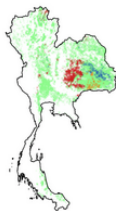
2016



2017



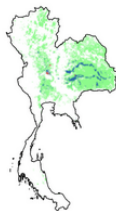
2018



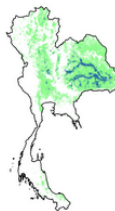
2019



2020



2021



2022

Evaluation Procedure

- We use leave one year out cross validation.
- In each CV fold, data from year y is used for testing. Data from $y' \neq y$ is used for insurance design (prediction model training and contract design).
- This gives us out of sample predictions for every observation in our data. We then compute utility-based performance metrics across all out of sample predictions.

Evaluation Details

- We use the training data and the same method to compute the premium for all methods.
- We modify Chen's method to use the same premium definition as ours.
- We use simulated data to estimate the CVaR when calculating the premium.
- We use a CRRA utility function with $\alpha = 1.5$ (Flatnes, Carter, and Mercovich (2018); Estefania-Salazar et al. (2024); Jensen, Stoeffler, et al. (2019))

- We focus on the Relative Insurance Benefit (RIB) (Kenduiwo et al. (2021); Estefania-Salazar et al. (2024)) as our main metric.
- Compares a proposed insurance contract to perfect insurance. It measures the welfare gains from perfect insurance that the proposed contract recovers.
- Interpretation: if RIB of an insurance J is 50%, we can think of it as meaning that it is 50% as good as perfect insurance. In other words, it yields half as many welfare gains as perfect insurance.

RIB Definition

Let w_i^J be the wealth of farmer i under insurance contract J . We denote perfect insurance with PI and no insurance with NI

$$CE(J) : u^{-1}(\mathbb{E}[u(w^J)])$$

$$\Delta CE(J) : = CE(J) - CE(NI)$$

$$RIB(J) : \frac{\Delta CE(J)}{\Delta CE(PI)}$$

$$\text{Utility : } u(w) = \frac{1}{1-\alpha} w^{\frac{1}{1-\alpha}}$$

Single Zone Evaluation Procedure

- Contract for each zone is designed as if the other zones didn't exist
- Required capital for zone z is based only on zone z 's contract.

$$\pi_z = \mathbb{E} \left[\sum_z I_z \right] + c_\kappa K$$
$$K_z = \text{CVaR}_{1-\epsilon}(I_z) - \mathbb{E}[I_z]$$

Single Zone Results

Method	RIB	Premium	Cost	Cost PI
VMX	0.679	0.014	0.017	0.027
Chantararat	-0.244	0.004	0.002	0.027
Chen	-2.676	0.073	0.062	0.027

Note on the value of c_κ

- Potential value for farmers is highly dependent on c_κ and risk aversion, α
- For the most commonly used value of α , farmers could only benefit from perfect insurance if $c_\kappa = 0.02$.
- As risk aversion increases, the gains from perfect insurance increase.

Effects of Cost of Capital and Risk Aversion

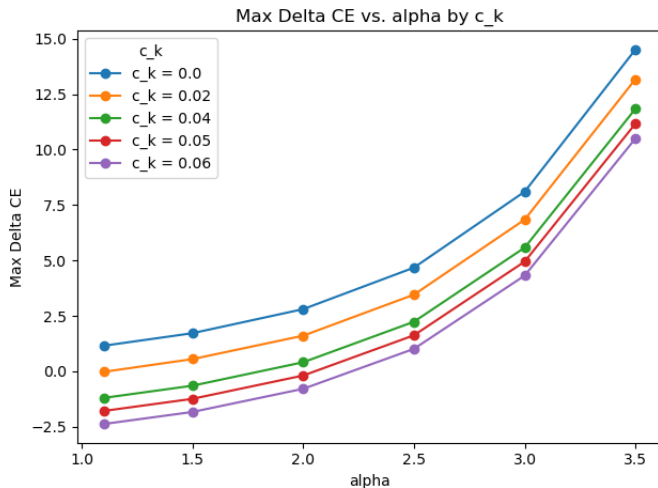


Figure: Perfect Insurance

General Insurance Behavior

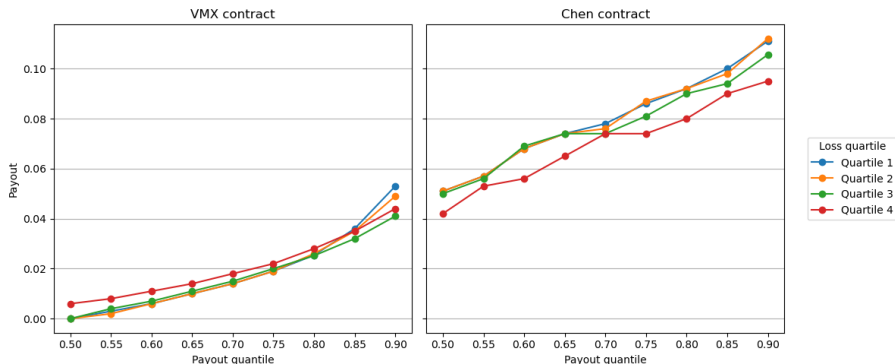
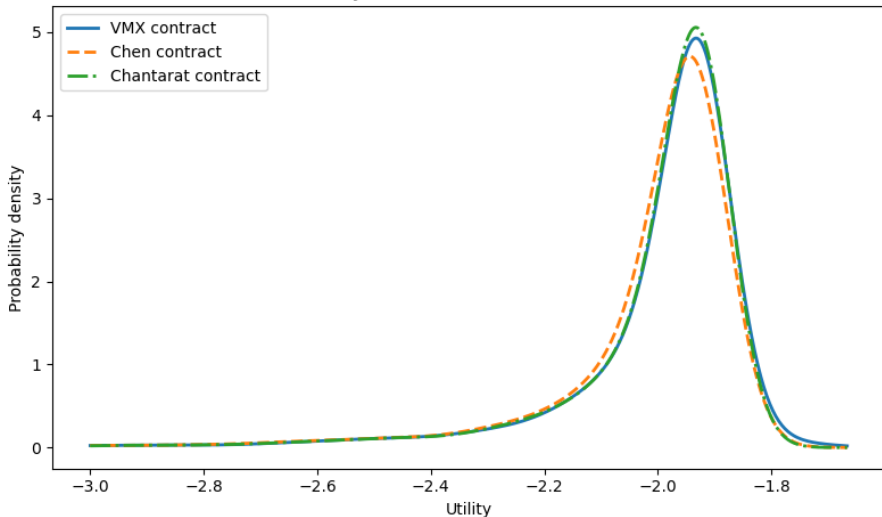


Figure: Each line shows the reverse CDF of payouts for a given loss quartile. Quartiles are numbered from smallest to largest, i.e. quartile 4 is for losses in the top quartile.

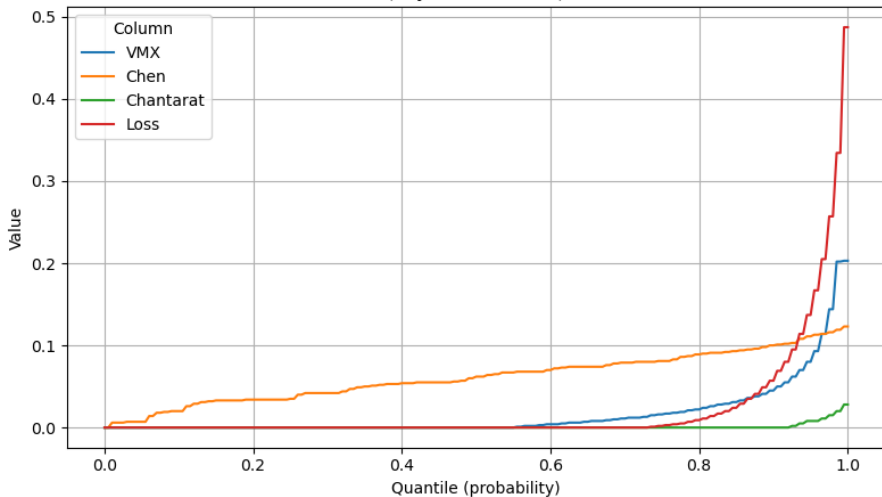
Outperforms in the in entire distribution

Utility distributions: VMX vs. Chen



Payout and Loss Quantile Functions

Empirical Quantile Function
(Payouts and Loss)



MZ vs SZ Case

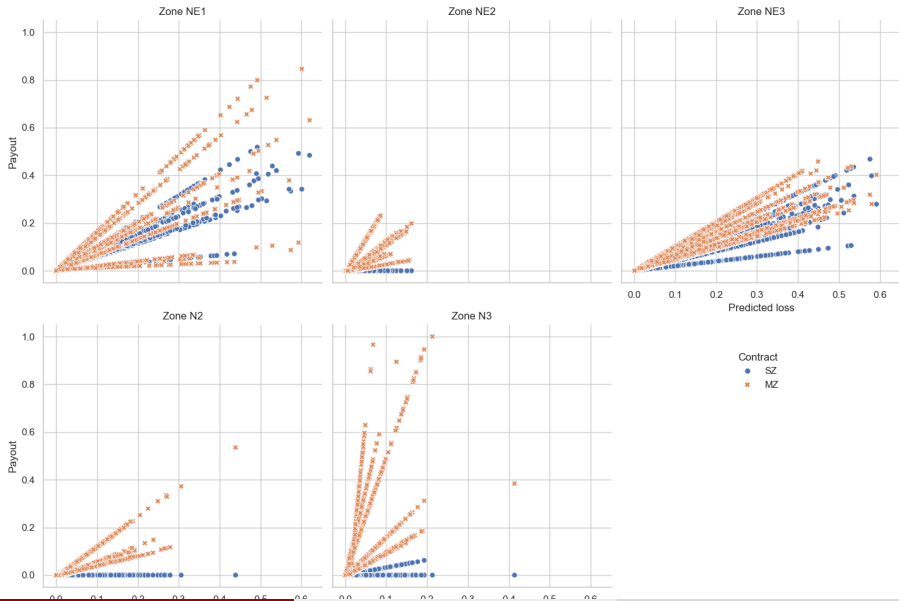
- Contracts can be designed independently (Chen and Baseline) or simultaneously (VMX).
- Required capital is calculated based on the portfolio of contracts

$$\pi_z = \mathbb{E} \left[\sum_z I_z \right] + c_\kappa \frac{\beta_z}{s_z} K$$
$$K = \text{CVaR}_{1-\epsilon} \left(\sum_z I_z \right) - \mathbb{E} \left[\sum_z I_z \right]$$

Multi zone results

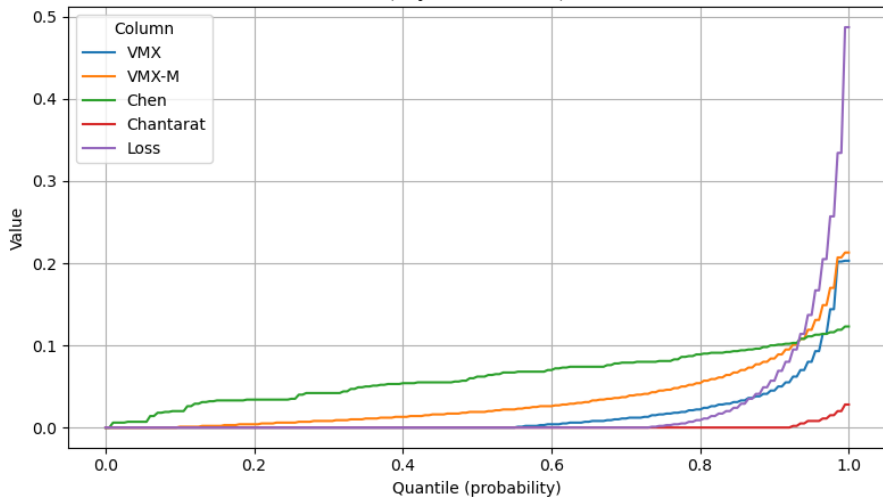
Method	RIB	Premium	Cost	Cost PI
VMX-M	0.592	0.035	0.036	0.027
VMX	0.524	0.014	0.017	0.027
Chantararat	-0.163	0.004	0.002	0.027
Chen	-1.909	0.074	0.062	0.027

SZ vz MZ Contracts

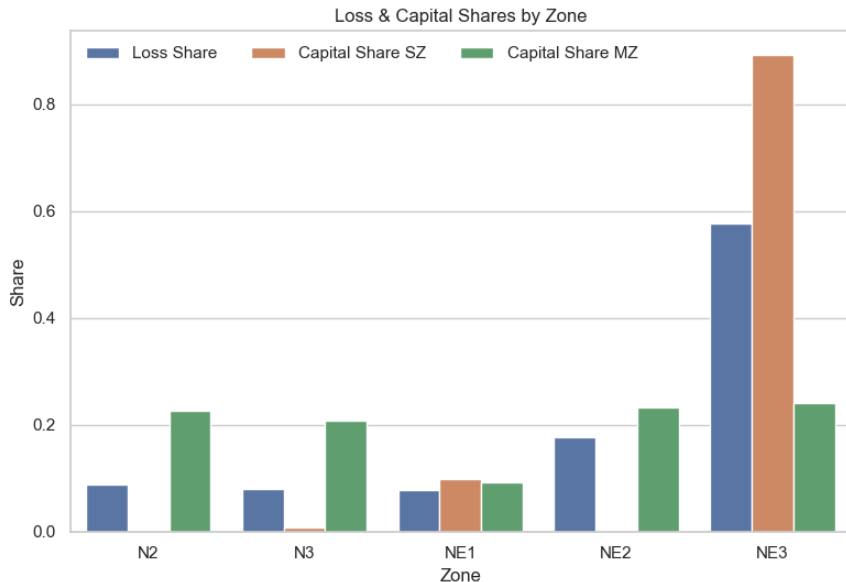


SZ vz MZ Payouts

Empirical Quantile Function
(Payouts and Loss)



Capital Allocation



Payout Precision: share of all payouts that are "valid".

$$P = \frac{\sum_{i:\hat{y}_i > 0} w_i \hat{y}_i \min\{\ell_i, \hat{y}_i\}}{\sum_{i:\hat{y}_i > 0} w_i \hat{y}_i}$$

Loss Recall: ratio of valid payouts to loss.

$$R = \frac{\sum_{i:\hat{y}_i > 0} w_i \hat{y}_i \min\{\ell_i, \hat{y}_i\}}{\sum_{i:\hat{y}_i > 0} w_i \ell_i}$$

Good Gains and Bad Losses

	Utility Gain	Utility Loss
Loss	✓	-
No Loss	-	✗

✓: utility gain during loss → "good gain"

✗: utility loss during loss → "bad loss"

Insurance Behavior Results

Method	RIB	Payout Precision	Loss Recall	Good Gain	Bad Loss
VMX-M	0.597	0.184	0.211	0.390	0.334
VMX	0.518	0.175	0.120	0.335	0.372
Chantararat	-0.222	0.328	0.010	0.616	0.514
Chen	-1.924	0.173	0.298	0.327	0.440

Summary of results

- Designing the contracts simultaneously leads to more aggressive contracts and larger utility gains.
- The program distributes capital costs more equitably, allowing more zones to be insured.
- The resulting contracts are more efficient.

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Idealized CVaR Model

- **Objective:** conditional value at risk of the farmers' loss net of insurance.
- **Constraint 1:** piecewise linear structure of the contract.
- **Constraint 2:** budget constraint.
- **Constraint 3:** definition of required capital.

$$\min_{a,b,\pi,K} \text{CVaR}_{1-\epsilon}(\ell - I(\theta))$$

$$\text{s.t. } I(\theta) = \min\{(a\hat{\ell}(\theta) + b)^+, P\} \quad (11)$$

$$\mathbb{E}[I(\theta)] + c_\kappa K \leq B \quad (12)$$

$$K = (\text{CVaR}_{1-\epsilon}(I(\theta)) - \mathbb{E}[I(\theta)]) \quad (13)$$

The problem is non-convex, so we need convex approximations

We use the following approximations of $I(\theta)$ to make the problem convex:

$$\overline{I(\theta)} \triangleq \max \left\{ 0, a\hat{\ell}(\theta) + b \right\}$$
$$\underline{I(\theta)} \triangleq \min \{ a\hat{\ell}(\theta) + b, K \}$$

- Note that $\overline{I(\theta)} \geq I(\theta)$ and $\overline{I(\theta)}$ is convex. Conversely, $\underline{I(\theta)} \leq I(\theta)$ and $\underline{I(\theta)}$ is concave.
- We replace $I(\theta)$ with either $\overline{I(\theta)}$ or $\underline{I(\theta)}$ where necessary to obtain conservative and convex approximations.
- We also need approximations or proxies for $E[I(\theta)]$ in constraint . We use $\pi_{SQ} = E[I_{SQ}(\theta)]$, where I_{SQ} is the contract designed using the status quo method, as a proxy for $E[I(\theta)]$ in constraint .